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MILITARY SPECIFICATION

COVERING, FLOOR, MILITARY AIRCRAFT, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for type, class and size of the floor covering for use in Military Aircraft.

1.2 Classification. Floor coverings shall be of the following types, classes and size as specified (see 6.2):

Type I - Vinyl impregnated fiberglass overlay.

Class 1 - Embossed overlay .031 inch nominal thickness.

Class 2 - Smooth overlay .024 inch nominal thickness.

Type II - Sponge underlay .25 inch nominal thickness with Type I overlay, Class 1 or 2, as specified.

Type III- Sponge underlay .50 inch nominal thickness with Type I overlay, Class 1 or 2, as specified.

Size A 50 inches wide X 150 inches long.

Size B 40 inches wide X 360 inches long.

Size C 30 inches wide X 360 inches long.

Size D 20 inches wide X 360 inches long.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1680

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SPECIFICATIONS

FEDERAL

CCC-T-191	Textile Test Methods.
QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet.
PPP-P-1136	Packaging of Coated (Plastic; Rubber) and Laminated Fabrics.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-Y-1140	Yarn, Cord, Sleeving, Cloth and Tape-Glass.
MIL-C-7514	Cloth Coated, Synthetic and Fibrous Glass.
MIL-P-15280	Plastic Foam, Unicellular, Sheet and Tubular Form, Elastomeric.
MIL-I-45607	Inspection Equipment, Supply and Maintenance; for Ordnance.
MIL-C-22787	Cloth, Coated, Fuel, and Flame Resistant.

STANDARDS

FEDERAL

FED-STD-595	Colors
FED-STD-601	Rubber: Sampling and Testing.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-293	Visual Inspection Guide for Cellular Rubber Items.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-45662	Calibration Systems Requirements.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions shall be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to a first article inspection (see 4.3 and 6.4).

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3.2 Materials. Materials shall be as specified herein, and when materials are used which are not specifically designated, they shall be entirely suitable for the purpose intended.

3.2.1 Materials shall be furnished as indicated on purchase orders.

3.2.2 Material requirements. The materials shall conform to the requirements outlined below.

3.2.3 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

3.3 Construction.

3.3.1 Type I.

3.3.1.1 Class 1. Embossed overlay material shall be vinyl coated fiberglass fabric per MIL-C-22787, type I. The cloth shall be per MIL-Y-1140, Form 4, No. 126, and shall have a thickness of .031 ± .003 inch. The wearing surface shall be embossed to create individual patterns consisting of 64 to 144 per square inch of surface area and having a height of .012 to .015 inch.

3.3.1.2 Class 2. Smooth overlay material shall be vinyl coated fiberglass fabric per MIL-C-22787, type I. The cloth shall be per MIL-Y-1140, Form 4, No. 126, and shall have a thickness of .024 ± .003 inch. The wearing surface shall be smooth.

3.3.2 Type II. The underlay material shall conform to the requirements of MIL-P-15280, type III. Overlay material shall conform to type I of this specification. The underlay and overlay materials shall be joined to form a unit structure. Adhesive cement conforming to requirements of MIL-P-15280, 3.7 shall be used for joining.

3.3.3 Type III. The underlay material shall conform to the requirements of MIL-P-15280, type III. Overlay material shall conform to type I of this specification. The underlay and overlay materials shall be joined to form a unit structure. Adhesive cement conforming to requirements of MIL-P-15280, 3.7 shall be used for joining.

3.3.4 Thickness. The thickness of the vinyl sponge underlay and the vinyl coated overlay, shall be determined in accordance with 4.5.10.

3.3.5 Flame resistance. The material shall be self extinguishing upon removal of the igniting flame when tested as specified in 4.5.1.

3.3.6 Compression deflection. The total laminate shall require a load of 5 ± 2 PSI to obtsin 25 percent compression when tested in accordance with 4.5.2.

3.3.7 Compression set. The total laminate shall now show compression set greater than 25 percent when determined in accordance with section 4.5.3.

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3.3.8 Water absorption. Water absorption of the vinyl sponge underlay material shall not exceed .1 pound per square foot of skin free surface when tested as specified in section 4.5.4.

3.3.9 Corrosion. The material shall not be corrosive when tested in accordance with section 4.5.5.

3.3.10 Tear resistance. The total laminate shall have a minimum tearing strength of 15 pounds in both the warp and filling directions of the impregnated fabric when tested in accordance with 4.5.7.

3.3.11 High temperature resistance. The material shall not show evidence of deterioration, tackiness, brittleness, or emit objectionable odor, and the shrinkage of either the length or width of the test specimen shall be less than 1 percent of the original length or width, respectively, when tested in accordance with 4.5.8.

3.3.12 Abrasion resistance. The total laminate shall have a minimum strength of 14 pounds in both warp and filling directions of the impregnated fabric after abrasion when tested in accordance with 4.5.9.

3.3.13 Adhesion. When tested in accordance with 4.5.12, the total laminate shall show no evidence of adhesive failure.

3.4 Weight. The weight of the laminated material shall be as follows when determined in accordance with 4.5.6:

Type II - 55 (max.) weight-oz/yd^2

Type III - 78 (max.) weight-oz/yd^2

3.5 Color. The color of the vinyl layer on the overlay shall be medium gray No. 36231 in accordance with FED-STD-595 when tested in accordance with 4.5.11. The color shall permeate the total thickness of the vinyl layer.

3.6 Identification of product. The underside of all coverings shall be marked per MIL-STD-130. The following entries shall be shown:

MFR's No. _____ Type _____ Class _____ Size _____

Order No. _____

Stock No. _____

Date of Manufacture _____

Name of Contractor _____

(The identification marking shall be in contrast to the vinyl sponge surface, arranged in two parallel rows approximately 16 inches apart, and repeated at intervals of 3 feet along the length of the covering(s). Character height shall be 3/8 inch minimum).

3.7 Workmanship. Covering shall be furnished in a thorough workmanlike manner and shall be free from defects that affect appearance or functionability. The quality of covering shall be as established by this specification.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

* a. First article test (see 4.3).

b. Acceptance tests (see 4.4).

* 4.3 First article testing.

* 4.3.1 First article test sample. The first article test sample for each class shall be of production quality and shall have a minimum width of 24 inches. The sample shall be of sufficient length to provide 14 square feet of material. The sample shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant, or at a laboratory selected by the contractor and approved by the procuring activity.

* 4.3.2 First article test report. The production test facility shall submit to the procuring activity a certified test report per MIL-STD-831 after completion of all first article testing. All test equipment shall meet the requirements of MIL-I-45607 and MIL-STD-45662.

* 4.3.3 First article tests. First article tests shall consist of all tests specified under 4.6 test methods.

4.4 Acceptance tests. Acceptance tests shall consist of:

a. Compression deflection per section 4.5.2.

b. Corrosion per section 4.5.5.

c. High temperature resistance per section 4.5.8.

d. Weight per section 4.5.6.

e. Color per section 4.5.11.

f. Flame resistance per section 4.5.1.

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4.4.1 Acceptance test material. Material used for performance of acceptance tests shall be taken from each batch of material produced to meet the requirements of each contract. Each batch produced shall be subjected to the acceptance tests as specified in 4.4 regardless of the number of batches required to meet the order requirements specified by the procuring activity.

4.4.2 Rejection. The failure of any portion of the acceptance tests shall constitute failure of the entire acceptance test requirements. Any batches on hand, or later procured, shall not be accepted until the extent and cause has been determined and the defects corrected by the contractor.

4.5 Test methods.

4.5.1 Flame resistance. A specimen of total laminate, approximately 3 inches wide by 12 inches long, shall be clamped in a suitable metal frame so that the exposed area for test purposes shall be at least 2 inches wide and 11 inches long. The specimen shall be clamped on three sides, leaving one end free, and at least 1/2 inch from the end of the frame for the application of the flame. The specimens shall be supported horizontally for the test, which shall be conducted in still air, or under an enclosed hood. The flame source shall be a Bunsen Burner having a 1.5 inch high, soft blue flame with no visible cone. Flame shall be applied to the free end of the specimen for 15 seconds and removed. The material shall then be observed for flame propagation.

4.5.2 Compression deflection. Three test specimens each, of type II and type III, shall be 1 $\pm$ .01 inch squares of the total laminate. The specimen thickness before testing shall be determined by gaging with apparatus specified in 4.6.10. The test apparatus shall be in accordance with FED-STD-601, method 12151, and shall be capable of measuring deflection in thousandths of an inch. The three specimens of each type shall be successively compressed by 25% of original thickness. The load applied at 25% deflection shall be recorded on each specimen. The average deflection value for each type specimen shall be reported as test values.

4.5.3 Compression set. Three test specimen each, of type II and type III, total laminate, 1.250 $\pm$ .005 inch diameter shall be prepared and thickness measured to nearest .001 inch using apparatus specified in 4.6.10. The compression device(s) shall consist of two or more flat plates with parallel faces, between which the specimens may be compressed. The compression device(s) shall be equipped with spacers of a length equal to the thickness the specimen are to be compressed. Bolts and nuts may be used to fasten the parallel plates against the spacers. (This device per FED-STD-601, method 12131). The specimen shall be compressed 25% of original thickness for 22 $\pm$ 2 hours at 75 $\pm$ 10°F. The specimen shall be removed from the device, and after a waiting period of 22 $\pm$ 2 hours, thickness measurements shall be taken. The compression set shall be calculated for each type specimen using the following formula and the average of three determinations reported as the test value.

$$\text{Compression set} = \frac{HO - H1}{HO - HS} \times 100$$

Where: HO = Original thickness - inches
H1 = Thickness after removal from device - inches
HS = Test compression thickness (spacers) - inches

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4.5.4 Water absorption. Three test specimen each, of type II and type III, total laminate, 4 X 4 inches square, measured to the nearest .01 inch, and weighed to the nearest .01 gram shall be immersed under a six inch head of water 48 ± 2 hours at 75 ± 10°F. The specimen shall then be removed from the water and air dried to remove visible surface water, and then reweighed. The weight shall be calculated as pounds of water per square foot of skin free surface for each type specimen and the average of three determinations reported.

4.5.5 Corrosion. A 6 inch X 6 inch specimen of the type II and type III, total laminate, shall be placed in a sterile container filled with a sufficient quantity of distilled water to cover the specimen. After soaking for 22 ± 2 hours at room temperature, the liquid shall be poured into a sterile evaporating pan containing a sterile coupon of bare aluminum sheet stock per QQ-A-250/4. The liquid shall be permitted to evaporate to dryness at a temperature not exceeding 158°F. A sample coupon of like aluminum shall be immersed in an equivalent amount of distilled water and the water evaporated in the same manner. Visible pitting of the test piece of aluminum in excess of any observed on the sample coupon shall be considered evidence of corrosion. Discoloration of the surface of the test sample coupons shall not be construed as corrosion.

4.5.6 Weight. A specimen of the total laminate, 6 inches X 6 inches square or larger, shall be measured to the nearest 0.1 inch and shall be weighed to the nearest .01 gram. The weight shall be reported as ounces per square yard.

4.5.7 Tear resistance. The tear resistance of the total laminate shall be determined in both the warp and filling directions of the impregnated fabric in accordance with the procedure specified in CCC-T-191, method 5136.

4.5.8 High temperature resistance. Specimens of the total laminate, at least 18 inches X 24 inches, shall be measured in both the length and the width to the nearest 0.1 inch and then subjected to an atmosphere of 155 ± 5°F for 60 ± 2 hours. The specimen shall be examined for evidence of deterioration, tackiness, brittleness, odor etc., and the length and width remeasured and the percentage of each calculated.

4.5.9 Abrasion resistance. The total laminate shall be tested in both the warp and filling directions of the impregnated fabric in accordance with the CCC-T-191, method 5304, with the exception that the specimen size shall be 3 inches X 9 inches. The specimen shall be held in the test device under a tension of 2 pounds and a load of 4 pounds. The wear surface shall be subjected to 2000 continuous cycles. The abradant shall be no. 1 1/2 emery cloth or equivalent. After the cycling period, a tear strength test specimen shall be cut from the abraded material and the bear strength determined as specified in method 5136, CCC-T-191. Five specimens in each the warp and filling directions shall be tested, and the average tear strength in both the warp and filling directions of the five specimens shall be reported.

4.5.10 Thickness. The total thickness of the material shall be determined by means of a dial gage graduated in thousandths of an inch, having a flat foot 1/4 inch in diameter exerting a total force of not more than 1/2 ounce. The thickness shall be measured at five positions and the average value reported as the total laminate thickness. The top surface (impregnated fabric) shall then be removed by freezing the sample and carefully peeling the top surface from the vinyl pad. The thickness of the surface material shall then be determined using the procedure and type equipment specified in method 5030 and 5100, CCC-T-191.

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4.5.11 Color. Visual comparison shall be conducted in accordance with the procedure contained in paragraph S5.3 and S6.2, FED-STD-595.

4.5.12 Adhesion. A 1" X 6" section of total laminate shall be cut from a sample. Tab ends shall be prepared by separating the fabric layer from the vinyl sponge at the bond line for a distance of one inch, using a sharp blade. The fabric and sponge tab ends shall then be pulled apart slowly by hand and the material observed during separation for evidence of adhesive failure.

4.6 Inspection of the preservation, packaging, and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PACKAGING

5.1 Preservation. Preservation, packaging and packing, shall be in accordance with applicable levels of PPP-P-1136C as specified.

5.2 Marking. In addition to any special marking required by the contract or order, interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Use. The floor coverings designated in this specification are intended for use in aircraft of all the services. The coverings are durable, easily maintained, impervious to most hydrocarbon and aromatic fluids, and have good acoustical qualities.

6.1.2 Type I. Type I covering is intended for use as an overlay for resilient materials for floor covering, but where requirements exist, this covering can be used singly for floor covering, wainscoting, panel covers, and other applications.

6.1.3 Type II and type III. Type II and III coverings are intended for use where durable, safe and resilient coverings are required. These types can be used with either smooth or embossed overlays conforming to type I, class 1 or 2. Class 1 overlay is for use where maximum wear conditions exist. Class 2 overlay is for use where minimum wear conditions exist, ie; under seats, storage areas etc. Installation of either type or class can be accomplished quickly and efficiently using minimum skills.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type, class and size required (see 1.2).

c. Quantity required.

d. Specific marking (see 3.7).

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* 6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD 1423) are invoked and the DD 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.3.2	First Article Test Reports	DI-T-5329	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, Volume II AMSDL. Copies of data item descriptions required by the contractors with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

* 6.4 First article. When a first article inspection is required, the item(s) should be (a first article sample), (a first production item, (a sample selected from the first production items) (or it may be a standard production item from the contractor's current inventory as specified in 4.1). The first article should consist of (one), (two), etc. unit(s). The contracting office should include specific instructions in acquisition documents upgrading arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspections to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

* 6.5 Subject term (key word) listing.

Floor covering

Mat rubber

Overlay floor

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made.

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